

NOMINATION LETTER – KHALID A. FAKHRO, PH.D. – ARAB AWARD IN GENETICS

Dear Esteemed Award Committee Members,

Last year, I submitted my self-nomination letter as a narrative of achievements over time (attached with supporting materials for reference). Given the 5 evaluation parameters provided this year, I have re-organized into relevant sections accordingly. My impact can be categorized at 3 levels: as Head of the Laboratory of Genomic Medicine (Academic), as Chief of Research at the largest women and children's hospital in Qatar (Executive leadership), and as an ecosystem-level architect of the nascent life sciences sector of the economy in Qatar (Innovator/Thought leader). Woven together, these pillars demonstrate a singular, highly cohesive contribution to the advancement of genomic science across Qatar and the wider Arab world.

Background Context and Commitment to the Region

In 2001, I graduated top of my country, and decided to pursue my passion for genetics – a field then perceived as offering *extremely limited* career prospects in the GCC. After completing my BA with Honors at the University of Chicago, my Masters and PhD at Yale University (receiving full scholarships throughout) and post-doctoral training at Weill-Cornell Medicine, I had to plan my next career move. Rather than remain within the established, well-funded scientific machinery of North America, I was driven by an aspirational vision: the next major scientific renaissance could, and *must*, emerge from the Arab world. This ambition led to the nascent Department of Genetic Medicine at WCMC-Qatar.

Ever since returning to the GCC, my mission has been twofold: to build world-class discovery science leveraging the unique genetic architecture and disorders affecting Arab populations that goes head to head with global efforts, and to architect local infrastructure to global standards, transforming our societies from knowledge consumers into global producers of knowledge and innovation. Over the years, I was lucky enough to be “building the plane while flying it,” using real-time learning to optimize my thinking and, in the process, make seminal contributions to the field of Genetics *from* the Arab world, *for* the Arab world. I am honored to summarize these for the Award committee as follows:

1. Research Activity: Driving Clinical and Paradigm-Shifting Discoveries

As a systems-level thinker, I have designed my lab's main projects as national programs, preserving two main themes: the ability to produce ground-breaking discoveries that help our patients *today*, and the readiness (through biobanking, umbrella consent, connection to EHR) for future studies using the technologies of *tomorrow*. This led to devising *highly translational* Programs which transform patient outcomes, while also creating sustainable infrastructure and talent for the Qatari life sciences sector. My lab's main research programs are as follows (with corresponding PMID references):

- **The Qatari Mendelian Program:** I established Qatar's **first** point-of-care Whole Exome, and later, Whole Genome Sequencing (WGS) operation (with clinical grade WGS rolled out in 2023, among the first in the region). By integrating genomics directly into the clinic as a primary diagnostic test, we achieved a remarkable ~60% overall diagnostic rate, discovering >100 novel pathogenic genes/variants, ending agonizing "diagnostic odysseys" for hundreds of families, and shifting the 'standard-of-care' from personalized, genome-guided interventions. Importantly, the Mendelian Program also focuses on hospital readiness and clinician-literacy, helping >20 clinics across the national health system better understand genomic medicine, appreciate technology strengths and limitations, and embed genomics-first thinking over the years. Finally, a critical part of the Program is our focus on mechanistic understanding via functional work in cellular models and Zebrafish. Today, the Mendelian Program has enrolled >4,200 participants and is transitioning to long-read sequencing and multi-omics integration – leveraging the broadly consented Mendelian biobank to increase diagnostic yield and produce real-world evidence to champion implementation of certain 'omics at the point of care in the coming years.

Key Papers: 31625567; 4032513; 4060177; 3848068; 3355968; 3460585; 3993786; 3442783; 3272693; 3194316; 4050802; 34795304

- **The BARAKA Study** (Building a National Research to Advance Knowledge in Autism; also, baraka means “blessing” in Arabic): Established the region’s **largest** bio-resource for Autism Spectrum Disorder (~2,000 sequenced, with fully consented biobank, EHR, and environmental questionnaires). Uncovered a regionally-specific paradigm shift – a significantly elevated burden of *recessive* Autism, contrary to the global consensus of sporadic/*de novo* origins. This fundamentally alters clinical management, enabling early screening and interventions. Further, leveraging the BARAKA Biobank, we introduced plasma proteomics and untargeted metabolomics to help boost diagnostic yield and inform treatment plans and are currently rolling out transcriptomics, stem cell models and AI-guided behavioral analysis as part of a multi-disciplinary “360” for each ASD patient (see Section 3).

Key Papers: 3636830; 3780553; 3529681; 3813575; 3266124; 34091591

- **Dissecting complex disease (via Mendelian extremes and multi-omics):** Leveraging consanguinity and population biobanks, my lab is making important strides studying the genetic basis of complex disorders in Qatar. For example, with male infertility—historically dismissed as multi-factorial with only few large-effect/causative genes—we leveraged familial clustering to discover novel genes causing spermatogenic failure, illuminating key functional pathways and demonstrating a monogenic basis at the ‘extremes’ of the distribution. This has now been replicated by ‘large’ global centers, making a major difference in the lives of affected couples. Similarly, in cancer, diabetes and metabolic disorders, we used multi-omics (including RNAseq, metabolomics, microRNAs, and epigenetics) to discover signatures predicting disease and complications. In one association study (the **largest** to date in any Arab population), we linked >1,300 untargeted metabolites to WES data in 997 Qataris, leveraging consanguinity to fine-map ‘sentinel’ SNPs to missense, causative variants. All metabolomic data and WES data were made available for the community, again a **first** for Arabs.

Key papers: 35150601; 3720256; 3691487; 2979087; 3473253; 2588019; 3686934; 4112633; 4118893; 3601211; 36659918

- **The NOOR Study for Preventative Newborn Genomic Screening:** Building on a series of high-impact papers we published mapping Mendelian burden, actionable variants and treatable inherited metabolic disorders in Qatar (*Genetics in Medicine* Editor’s Choice, Oct 2024), we designed the NOOR Study (Newborn Optimized multi-omic Screening; also, noor means “light” in Arabic) in 2023 as the **first** national newborn genomic screening program in the GCC. Our program has since grown in collaboration with BeginNGS, and all patients are consented for data sharing, future use, and multi-omics. We envision this will be the first ancestrally diverse, newborn biobank in the world, providing valuable references for this age group globally.

Key papers: 3362300; 3928696; 3632947; 3504641; 3516915; 3894274; 3902006; 42161750

- **A Bedrock of Foundational Population Genomics Studies:** Though the Hamdan Award focuses on medical genetics/genomics, much of the work done on patients in our region would be *incomplete* without adequate understanding of population genetics and background variation. My lab has contributed significantly to this space, making numerous breakthroughs and publishing internationally acclaimed population studies. Some highlights as follows:

- **2016:** Published the **first** genomic dissection of Arab population structure (cover of *Genome Research*), proving deep, ancient allelic diversity of the Middle East.
- **2017:** Published the **first** CNV map of any Arab population, demonstrating higher levels of homozygous deletions in Gulf Arabs, priming the search for natural human knockouts.
- **2019:** Published the **first** study of reference bias on Arab genome interpretation, with ~15% of ‘variant alleles’ shown to be ‘reference’ in Qataris (MAF>50%), including ~200 mis-labelled ‘pathogenic’ in OMIM. Published **first** Qatari ref genome (QTRG), and contributed >1,000 sequences to SRA (the **largest** Arab genome submission to a public database to date)

- **2021:** Published the **first** high-resolution map of population structure of any Arab population (6,218 Qatari genomes). Discovered 5 ancestral sub-populations, showing inward migration into Arabia from neighboring continents, and detecting outward gene flow into Mediterranean Europe and as far as South America. Provided imputation reference panel for 12,432 haplo-types, including the largest collection ChrY J1a2B haplogroups (29 across 1,491 individuals).
- **2022:** Demonstrated utility of the **first** customized, population-specific microarray (Qchip1) for high-throughput genotyping and detection of pathogenic + pharmacogenomic variants in the Qatari population. Accredited for adoption in clinical settings (newborn/premarital screening).
- **2024:** Published the **first** set of long-read reference genomes for five MENA populations (cover of *Nature Genetics*), uncovering >42 megabases of entirely novel sequences and serving as a pioneering global reference for these underrepresented populations.
- **2026:** Published the **largest** study on structural variants in Arabs, cataloging deletions, duplications, inversions, translocations and tandem repeats (>150k variants in ~6100 individuals). Leveraged consanguinity to discover >180 completely knocked out genes, and >50 'essential' genes. Discovered multiple non-coding loci with large effect on extreme phenotypic and biobank traits, and demonstrated functional impact using proteomics.

Key papers: 2936236; 3464233; 4153013; 4178154; 3990101; 3858427; 3507213; 3786804; 2672871; 3342006; 33563995

2. Qualitative and Quantitative Assessments of Publications

High-quality papers are the central currency of our field, and building a reputation for excellence from the Middle East is not a trivial ask. While many universities use number of papers as a productivity metric, I have always prioritized impact/quality over quantity, not only for my lab, but for the wider Sidra Research Branch. Our labs are not judged by volume of papers, but by IF and impact on patient care:

- **Qualitative Impact:** As detailed in previous sections, many of my papers report discoveries of new disease genes or biomarkers, creating foundational databases for Arab genomic medicine, and impacting implementation at a national level. Further, several of our studies have earned prestigious accolades, including being featured on journal covers (*Genome Research*, *Nature Medicine*, *Nature Genetics*, etc.), global media coverage (see Section 5), and being highlighted by journals, e.g. *Genetics in Medicine* Editor's Choice. Importantly, our landmark papers contain releases relevant for genomics of Arab populations, including tools (code), imputation panels, and primary data (100 ASD families deposited in MSSNG, >1,300 Arab exomes/genomes on SRA, ~1,000 untargeted metabolomics profiles on Arab ancestries, etc.)
- **Quantitative Track Record:** I have authored 107 articles, high-impact reviews, and book chapters (details in Supporting Material). My cumulative Impact Factor is ~974.5, giving a mean IF of 9.1. Over 85% of my lab's papers are in Q1 journals, and 20% are in highly prestigious *Nature Index* journals, including *Nature* (and *Nat Med/Genet/Comms*), *Cell*, *Science Transl Med*, *PNAS*, etc.

3. Collaboration and Interdisciplinarity

I believe that real innovation occurs at the intersection of fields, and this is one of the major reasons why I chose to build my career in Education City, where I'm linked to a truly multi-disciplinary environment. The following are my main collaborations and inter-disciplinary efforts:

- **BARAKA '360-Degree' Autism Framework:** I was recently appointed to lead the Research Pillar of the National Autism Plan, and I am transforming BARAKA from a two-dimensional study (genetics x environment) to a full '360-degree' model for each patient. Combining expertise from local universities/institutes in social sciences, AI, multi-omics, stem cell modeling, and wearables, we are building the region's **first** truly personalized, interdisciplinary research framework for Autism.
- **AI & Computational Biology:** Partnered with Qatar Computing Research Institute (QCRI) to deploy machine and deep learning architectures to predict variant pathogenicity.

- **Islamic Bioethics:** Partnered with HBKU's CILE on several reports/papers on *Genomics and Islamic Ethics*, establishing global baseline for culturally compliant precision medicine.
- **Science Journalism:** I initiated a **first-of-its-kind** partnership with Northwestern University in Qatar, embedding media students within Sidra labs to translate complex molecular concepts (like advanced therapies and precision medicine) into accessible public media.
- **Health Economics:** Initiated a health economics study with Qatar Foundation to evaluate the utility of newborn genomic screening, the result of which is leading to roll-out of gNBS across Qatar.
- **Elite International Alliances:** Many of the programs and papers I described in previous sections came from collaborative projects with premier global institutions, including King's College London, UCL, Baylor College of Medicine, Genomics England, SickKids, CHOP and GOSH. In architecting such high quality collaborations, my lab has received >\$15M in competitive grant funding, which allowed us to build substantial infrastructure and train a whole generation of scientists at these global institutions, building lasting capacity for Qatar.

4. Ancillary Activities: Ecosystem Building & Senior Leadership

In addition to building programs that aim to lift the ecosystem as a whole, I have contributed to building national capacity for genomic medicine on a system-wide level. Building off a strong foundation in academic research, I managed to transform our hospital into a research-hospital where genome-guided care continues to roll out to new clinics. Most recently, I began to integrate our efforts with the nascent innovation economy, striving to build a vibrant life sciences sector. Relevant highlights include:

- **Translational Infrastructure:** I directed the delivery of Qatar's **first** clinical-grade WES and WGS services, eliminating the need for sample send-outs and decreasing turnaround times for diagnostics. I also built Qatar's **first** fully accredited, clinical-grade GMP facility, enabling for the first time a pediatric bone marrow transplant to occur in-country, and shifting our capabilities to actively manufacture cures in the coming years. Today, Sidra is known for its world-class infrastructure, with outstanding core facilities including microscopy, flow cytometry, mass spec, zebrafish, microbiome and stem cell cores, all of which were developed to transform patient-care through research. We will soon be producing local genetic medicines, closing the loop for rare disease.
- **Forging the Innovation Economy:** To commercialize biomedical discoveries, I established Qatar's first healthcare-specific *Innovation Lab*, formally linking start-ups and private companies to our world-class infrastructure, exceptional staff, and extensive patient cohorts. I signed partnership agreements with several Venture Capital firms and start-ups, and successfully launched Qatar's first Middle East Healthcare Accelerator (MEHA) with Deerfield capital.
- **Human Capital and Talent Development:** I was deeply involved in setting up the country's first graduate programs at the College of Health and Life Science (HBKU). I created the syllabus for the Genomics and Precision Medicine course, taught a large fraction of classes in the first two years, and supervised and graduated Qatar's **first** PhD student. Since then, I have supervised >20 graduate students, and remain the only external member on the HBKU Graduate Program Committee (fourth-consecutive term). Moreover, through my strategic direction, institutional policies, and direct oversight, our Branch at Sidra Medicine has supported > 250 global investigators, students and trainees over the years, many from neighboring Arab countries.
- **Senior National Appointments:** As a recognized leader in regional healthcare strategy, I hold critical executive appointments across the national landscape and have represented Qatar nationally (including at UNGA), including Board Member of the Qatar Precision Health Institute (QPHI), Vice Chair of the Board at Diabetes Qatar, Chair of the Research Pillar on the National Autism Planning Committee, and a member of the QF's Technical Executive Committee for Precision Health. I also serve on multiple advisory boards, including the Healthcare Management Program for University of Doha for Science and Technology, the HBKU Graduate Program Committee, and the Advisory Board for the International School for Medical Sciences and Engineering. Finally, I hold academic

appointments at HBKU and at WCM-Q (Dept of Genetic Medicine) as well as the WCM-NY graduate school (Dept of Neuroscience).

5. Communication, Dissemination, and Impact on Society

Over the years, I've built several strategic campaigns to improve communication and ensure dissemination of the impact of our translational research on society:

- **Sidra Conferences and Workshops:** I am the executive sponsor of Sidra's conferences and workshops arm, focusing on delivering 4-5 high quality events every calendar year. In addition to our flagship PMFG (see below) we run 3-4 'thematic' conferences and workshops every year, spanning topics such as genome assemblies, zebrafish models, stem cell and advanced therapies, etc. Accompanying workshops attract trainees from the region, ensuring they get access to the latest technologies in world-class labs under exceptionally tailored mentorship.
- **PMFG:** Launched the region's longest-standing genomics conference. Now in its 12th year, the **Precision Medicine and the Future of Genomics (PMFG)** summit has become globally recognized for its high quality and curated content, bringing international thought leaders, public figures, senior stakeholders and the entire scientific and life sciences sector together. While other countries pay millions to attract 'global' conferences to the GCC, PMFG is entirely home-grown, built to global standards over years of personal outreach, networking and perseverance.
- **Global Media Positioning:** I routinely leverage high-profile local and international television interviews to challenge legacy narratives, explicitly positioning the GCC as the premier global home and active laboratory for advanced genetic medicine (see Supporting Materials for links)
- **Science Journalism:** See section 3 above.
- **Public Engagement:** We launched a highly successful national Science Café, allowing citizens to interact seamlessly with researchers in an unassuming local setting (Msheireb museums), and am currently launching a patient-advisory group for research. In addition, I strive to name my research studies with acronyms that represent Arabic words that help capture local imagination and destigmatize research to the general public, e.g. BARAKA and NOOR (as described in Section 2 above; for more examples, see the Grants section in my CV).

Conclusion

Over the years, I chose a road less travelled, returning to a 'blank slate' in the GCC rather than enjoying a pre-built career path in North America or Europe. Today, seeing our "Qatari experience" actively utilized as the gold-standard benchmark for launching national genome programs across the Middle East brings me a sense of pride, and seeing our trainees graduate to lead their own labs and make life-changing discoveries for patients brings its own profound sense of achievement.

My career demonstrates an unwavering commitment to the field of genetics and genomic medicine and to patients and families across the Arab world. Through pioneering research that drives paradigm shifts, construction of world-class infrastructure, and a consistent pursuit of excellence, my mission remains steadfast: to ensure the Arab world leads the global future of genomic medicine. Doing so as a Hamdan Foundation Award recipient would, of course, be the ultimate honor.

Sincerely,



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